

Vera AI Challenge — Submission Brief

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Date: 19 September 2026

Challenge: magicpin Vera AI Challenge

1. Public bot URL (for evaluation)

<https://vera.65.20.84.111.sslip.io>

Check	Endpoint
Liveness	GET /v1/healthz
Team / approach	GET /v1/metadata
Interactive docs	GET /docs

The service is deployed on a cloud VPS and will remain online for the evaluation window.

2. What I built

A **stateful HTTP bot** that implements Magicpin's Vera message engine:

- Ingests the **4-context framework** (Category, Merchant, Trigger, Customer)
 - Composes grounded WhatsApp-style messages with a clear CTA and rationale
 - Handles multi-turn merchant replies (send / wait / end)
 - Exposes the full judge contract: /v1/context , /v1/tick , /v1/reply , /v1/healthz , /v1/metadata
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3. Approach (how it thinks)

Message composition

- **Trigger-kind dispatch** — different framing for research digest, recall, perf dip, festival, competitor, IPL, etc.
- **Grounded in context only** — numbers, offers, digest titles, slots, and names come from pushed JSON (no invented competitors or fake offers)

- **Category + merchant fit** — peer CTR, active offers, owner first name, Hinglish when `hi` is in languages
- **Restraint** — suppression keys, expiry checks, one proactive action per merchant per tick; empty `actions: []` when nothing is worth sending

Multi-turn conversation quality

Scenario	Behaviour
WhatsApp Business auto-reply	Detect canned patterns → exit quickly (don't burn turns)
Affirmative intent ("ok let's do it", "haan")	Switch qualifying → action ; deliver concrete next step
Hostile / "stop" / not interested	Graceful <code>end</code>
Off-topic (e.g. GST)	Stay on-mission politely

Stack

- **Python + FastAPI + Uvicorn**
- Deterministic composer (reliable under 30s timeouts)
- Optional LLM polish via Groq (can be enabled without changing the API contract)
- Docker-ready; currently running on a persistent VPS (`sslip.io`)

4. Design principles aligned with Vera's pain points

From the challenge brief, production Vera struggles with auto-reply pollution, intent handoff failures, and generic copy. This submission targets those directly:

1. **Faster auto-reply exit** — don't waste 2–3 turns
2. **Immediate action on clear intent** — no re-qualification after "let's do it"
3. **Service+price specificity** — e.g. Cleaning @ ₹299, not vague "10% off"
4. **Curiosity / reciprocity levers** where the trigger supports them (research digests, curious-ask)

5. Example of output quality

Weak (generic):

"Hi Doctor, want to run a discount campaign today?"

This bot (grounded):

Uses merchant CTR vs peer median, live catalog offers, digest source/page, customer slots and language preference — then asks one low-friction next step.

6. Code & reproducibility

- Private GitHub repository with full source, Dockerfile, and documentation
 - Repo: `https://github.com/lovepreet166/vera-ai-challenge` (collaborator access available on request)
 - Local self-test via Magicpin's `judge_simulator.py` contract
 - Expanded dataset generated from the official seed pack
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7. Operational readiness

Item	Status
Public HTTPS URL	Live
Health checks	Passing
Team metadata	Lovepreet / Lovepreet Singh
Timeout-safe responses	Designed for $\leq 30s$
Judging window uptime	VPS kept running

8. Closing

I'm excited about Vera's merchant-growth problem space and happy to walk through architecture, trade-offs, or a live trace of `/v1/tick` + `/v1/reply` during review.

Thank you for the opportunity.

— Lovepreet Singh

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